Chapter 17 Review

Campbell Biology, AP Edition, 12e

Chapter 17 Review

Summary of Key Concepts

Given that polypeptides are the link between genotype and phenotype, how could this fact be used to support the hypothesis that RNA, not DNA, evolved first, sometimes referred to as the RNA World hypothesis? **(BIG IDEAS 1 & 4)**

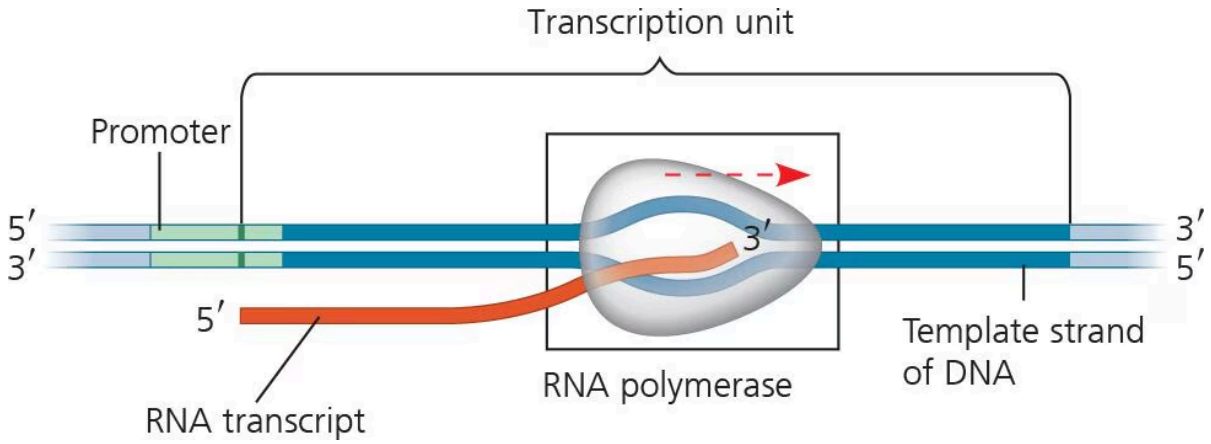
Concept 17.1: Genes specify proteins via transcription and translation

- Beadle and Tatum's studies of mutant strains of ***Neurospora*** led to the one gene–one polypeptide hypothesis. During **gene expression**, the information encoded in genes is used to make specific polypeptide chains (enzymes and other proteins) or RNA molecules.
- **Transcription** is the synthesis of RNA complementary to a **template strand** of DNA. **Translation** is the synthesis of a polypeptide whose amino acid sequence is specified by the nucleotide sequence in **messenger RNA (mRNA)**.
- Genetic information is encoded as a sequence of nonoverlapping nucleotide triplets, or **codons**. A codon in mRNA either is translated into an amino acid (61 of the 64 codons) or serves as a stop signal (3 codons). Codons must be read in the correct **reading frame**.

Describe the process of gene expression, by which a gene affects the phenotype of an organism.

Concept 17.2: Transcription is the DNA-directed synthesis of RNA: *A Closer Look*

- RNA synthesis is catalyzed by **RNA polymerase**, which links together RNA nucleotides complementary to a DNA template strand. Transcription follows the same base-pairing rules as DNA replication, except that in RNA, uracil (U) substitutes for thymine (T).

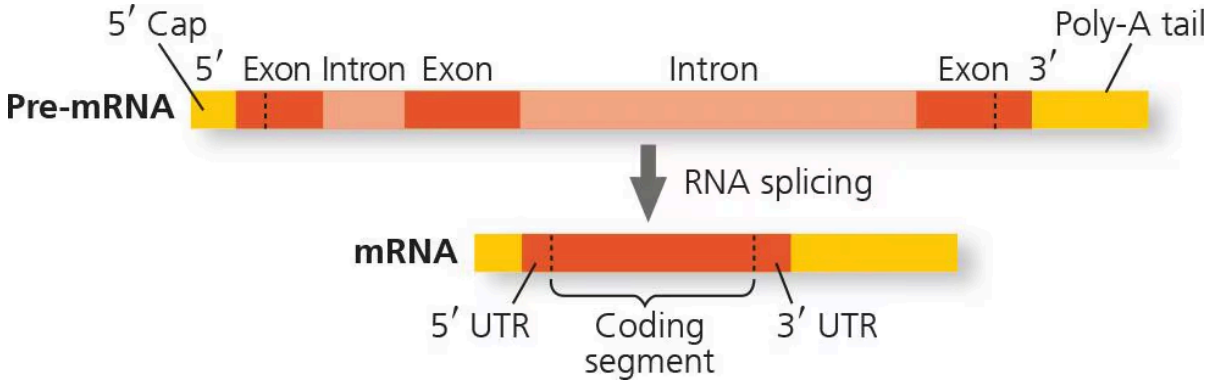


- The three stages of transcription are initiation, elongation, and termination. A **promoter** sequence, often including a **TATA box** in eukaryotes, establishes where RNA synthesis is initiated. **Transcription factors** help eukaryotic RNA polymerase recognize promoter sequences, forming a **transcription initiation complex**. Termination differs in bacteria and eukaryotes.

Compare the initiation of transcription in bacteria and eukaryotes.

Concept 17.3: Eukaryotic cells modify RNA after transcription

- Eukaryotic mRNAs undergo **RNA processing**, which includes RNA splicing, the addition of a modified nucleotide 5′ **cap** to the 5′ end, and the addition of a **poly-A tail** to the 3′ end. The processed mRNA includes an untranslated region (5′ UTR or 3′ UTR) at each end of the coding segment.
- Most eukaryotic genes are split into segments: They have **introns** interspersed among the **exons** (the regions included in the mRNA). In **RNA splicing**, introns are removed and exons joined. RNA splicing is typically carried out by **spliceosomes**, but in some cases, RNA alone catalyzes its own splicing. The properties of RNA allow some RNAs (called **ribozymes**) to act as catalysts. The presence of introns allows for **alternative RNA splicing**.



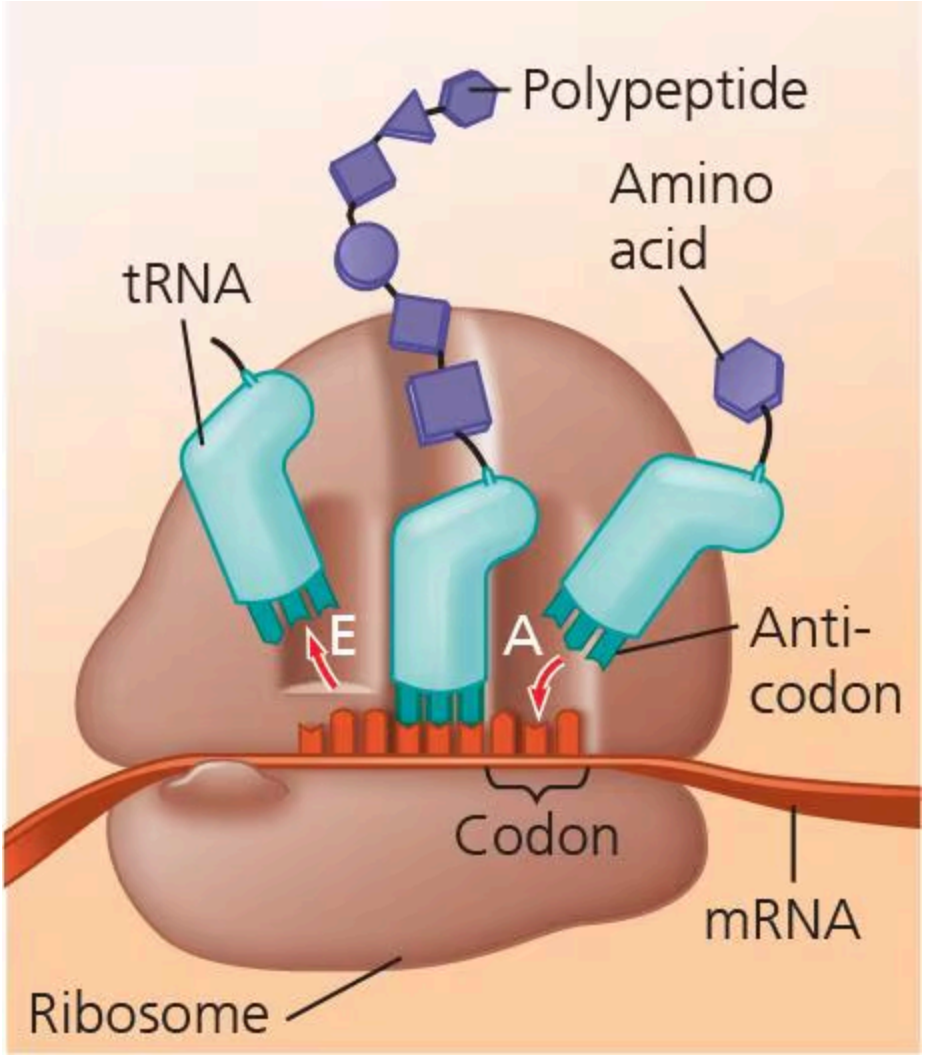
Overview: In the nucleus, transcription of D N A results in pre-m R N A. R N A processing leads to m R N A, which leaves the nucleus. Translation of the m R N A takes place in the cytoplasm on ribosomes, where polypeptides are synthesized. Pre-m R N A labeled as follows: 5 prime cap; 5 prime U T R; Exon (codon numbers 1-30); short intron; exon (codon numbers 31-104); long intron; exon (codon numbers 105-146); 3 prime U T R; poly-A tail. Arrows flow to result after introns cut out and exons spliced together. m R N A: 5 prime cap; 5 prime U T R; Coding segment 1-146; 3 prime U T R; poly-A tail.

What function do the 5′ *cap and the poly-A tail serve on a eukaryotic mRNA?*

Concept 17.4: Translation is the RNA-directed synthesis of a polypeptide: *A Closer Look*

- A cell translates an mRNA message into protein using **transfer RNAs (tRNAs)**. After being bound to a specific amino acid by an **aminoacyl-tRNA synthetase**, a tRNA lines up via its **anticodon** at the complementary codon on mRNA. A ribosome, made up of **ribosomal RNAs (rRNAs)** and proteins, facilitates this coupling with binding sites for mRNA and tRNA.
- Ribosomes coordinate the three stages of translation: initiation, elongation, and termination. The formation of peptide bonds between amino acids is catalyzed by rRNAs as tRNAs move through the

A and **P sites** and exit through the **E site**.



Overview: In the nucleus, transcription of D N A results in pre-m R N A. R N A processing leads to m R N A, which leaves the nucleus. Translation of the m R N A takes place in the cytoplasm on ribosomes, where polypeptides are synthesized. Messenger R N A is inside ribosome. Various amino acids are drawn present in the cytoplasm and some are already attached to different t R N A molecules. Here, t R N As are drawn as upside-down L-shaped molecules with triplet anticodons on one end and attachment points for amino acids on the other end.

- After translation, during protein processing, proteins may be modified by cleavage or by attachment of sugars, lipids, phosphates, or other chemical groups.
- Free ribosomes in the cytosol initiate synthesis of all proteins, but proteins with a **signal peptide** are synthesized on the ER.
- A gene can be transcribed by multiple RNA polymerases simultaneously. Also, a single mRNA molecule can be translated simultaneously by a number of ribosomes, forming a **polyribosome**.

In bacteria, these processes are coupled, but in eukaryotes they are separated in space and time by the nuclear membrane.

How do tRNAs function in the context of the ribosome during translation?

Concept 17.5: Mutations of one or a few nucleotides can affect protein structure and function

- Small-scale **mutations** include **point mutations**, changes in one DNA nucleotide pair, which may lead to production of nonfunctional proteins. **Nucleotide-pair substitutions** can cause **missense** or **nonsense mutations**. Nucleotide-pair **insertions** or **deletions** may produce **frameshift mutations**.
- Spontaneous mutations can occur during DNA replication, recombination, or repair. Chemical and physical **mutagens** cause DNA damage that can alter genes.
- The **CRISPR-Cas9 system** is a powerful new **gene-editing** technique with the potential to correct genetic mutations that cause disease. Significant technical and ethical questions have been raised about how this could or should be used.

What will be the results of chemically modifying one nucleotide base of a gene? What role is played by DNA repair systems in the cell?